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SQL Basics Cheat SheetSQL , or Structured Query Language , is a language to talk to databases. It allows you to select specific data and to build

s. Today, SQL is a universal language of data. It is used in practically all technologies that

SAMPLE DATA

CITY

id name country_id population rating

1 Paris 1 2243000 5

2 Berlin 2 3460000 3

... ...COUNTRY

id name population area

1 France 66600000 640680

2 Germany 80700000 357000

... ...

ALIASES

COLUMNS

```
SELECT name AS city_name
FROM city;
```

TABLES

```
SELECT co.name, ci.name
FROM city AS ci
JOIN country AS co
ON ci.country_id = co.id;
```

QUERYING MULTIPLE TABLES

INNER JOIN

```
SELECT city.name, country.name
```

FROM city

[INNER] JOIN country

ON city.country_id = country.id;

CITY

id name country_id

1 Paris 1

2 Berlin 2

3 Warsaw 4COUNTRY

id name

1 France

2Germany

3IcelandJOIN (or explicitly INNER JOIN) returns rows that have matching values in both tables.

LEFT JOIN

SELECT city.name, country.name

FROM city

LEFT JOIN country

ON city.country_id = country.id;

CITY

id name country_id

1 Paris 1

2 Berlin 2

3 Warsaw 4COUNTRY

id name

1 France

2Germany

NULL NULLLEFT JOIN returns all rows from the left table with corresponding rows from the right table. If there's no matching row, NULL s are returned as values from the second table.

RIGHT JOIN

SELECT city.name, country.name

FROM city

RIGHT JOIN country

ON city.country_id = country.id;

CITY

id name country_id

1 Paris 1

2 Berlin 2

NULL NULL NULLCOUNTRY

id name

1 France

2Germany

3IcelandRIGHT JOIN returns all rows from the right table with corresponding rows from the left table. If there's no matching row, NULL s are returned as values from the left table.FULL JOIN

SELECT city.name, country.name

FROM city

FULL [OUTER] JOIN country

ON city.country_id = country.id;

CITY

id name country_id

1 Paris 1

2 Berlin 2

3 Warsaw 4

NULL NULL NULLCOUNTRY

id name

1 France

2Germany

NULL NULL

3IcelandFULL JOIN (or explicitly FULL OUTER JOIN) returns all rows from both tables if there's no matching row in the second table, NULL s are returned.

CITY

country_id id name

6 6San Marino

7 7Vatican City

5 9 Greece

10 11 MonacoCOUNTRY

name id

San Marino 6

Vatican City 7

Greece 9

Monaco 10NATURAL JOIN

SELECT city.name, country.name

FROM city

NATURAL JOIN country;NATURAL JOIN will join tables by all columns with the same

name.

NATURAL JOIN used these columns to match rows:

city.id, city.name, country.id, country.name

NATURAL JOIN is very rarely used in practice.CROSS JOIN

SELECT city.name, country.name

FROM city

CROSS JOIN country;

SELECT city.name, country.name

FROM city, country;CROSS JOIN returns all possible combinations of rows from both tables. There are two syntaxes available.

CITY

id name country_id

1 Paris 1

1 Paris 1

2 Berlin 2

2 Berlin 2COUNTRY

id name

1 France

2Germany

1 France

2GermanyQUERYING SINGLE TABLE

Fetch all columns from the country table:

SELECT *

FROM country;

Fetch id and name columns from the city table:

SELECT id, name

FROM city;

SELECT name

FROM city

ORDER BY rating DESC ;Fetch city names sorted by the rating column

in the DESC ending order:SELECT name

FROM city

ORDER BY rating [ASC] ;Fetch city names sorted by the rating column

in the default ASC ending order:SELECT name

FROM city

WHERE name LIKE '_ublin';Fetch names of cities that start with any letter followed by

'ublin' (like Dublin in Ireland or Lublin in Poland):SELECT name

FROM city

WHERE name != 'Berlin'

AND name != 'Madrid';Fetch names of cities that are neither Berlin nor Madrid:

SELECT name

FROM city

WHERE rating IS NOT NULL ;Fetch names of cities that don't miss a rating value:

SELECT name

FROM city

y_id IN (1, 4, 7, 8);Fetch names of cities that are in countries with IDs 1, 4, 7, or 8:FILTER

SELECT name

FROM city

HERE rating > 3;Fetch names of cities that have a rating above 3:COMPARISON OPERA

SELECT name

FROM city

WHERE name LIKE 'P%'

R name LIKE '%s';Fetch names of cities that start with a 'P' or end with an 's':TEXT OPER

SELECT name

FROM city

E population BETWEEN 500000 AND 5000000;Fetch names of cities that have a populati

500K and 5M:OTHER OPERATORS

avg(expr) average value for rows within the group

count(expr) count of values for rows within the group

max(expr) maximum value within the group

min(expr) minimum value within the group

sum(expr) sum of values within the groupAGGREGATE FUNCTIONSCYCLING

idnamecountry

1 YK DE

2 ZG DE

3 WT PL

... ...SKATING

idnamecountry

1 YK DE

2 DF DE

3 AK PL

... ...AGGREGATION AND GROUPING

GROUP BY groups together rows that have the same values in specified columns.

It computes summaries (aggregates) for each unique combination of values. SUBQUERI

t is nested inside another query, or inside another subquery.There are different types of su

o a single result. The combined queries must return the same number of columns and com

CITY

country_id count

1 3

2 3

4 2CITY

id name country_id

1 Paris 1

101Marseille 1

102 Lyon 1

2 Berlin 2

103 Hamburg 2

104 Munich 2

3Warsaw 4

105 Cracow 4

EXAMPLE QUERIES

SELECT COUNT(*)

FROM city;Find out the number of cities:

SELECT COUNT(rating)

FROM city;Find out the number of cities with non-null ratings:

SELECT COUNT(DISTINCT country_id)

FROM city;Find out the number of distinctive country values:

SELECT MIN(population), MAX(population)

FROM country;Find out the smallest and the greatest country populations:

SELECT country_id, SUM(population)

FROM city

GROUP BY country_id;Find out the total population of cities in respective countries:

SELECT country_id, AVG(rating)

FROM city

GROUP BY country_id

(rating) > 3.0;Find out the average rating for cities in respective countries if the average is

SELECT name

FROM cycling

WHERE country = 'DE'

UNION / UNION ALL

SELECT name

FROM skating

WHERE country = 'DE';UNION combines the results of two result sets and removes duplic

UNION ALL doesn't remove duplicate rows.

This query displays German cyclists together with German skaters:

INTERSECT

SELECT name

FROM cycling

WHERE country = 'DE'

INTERSECT

SELECT name

FROM skating

WHERE country = 'DE';INTERSECT returns only rows that appear in both result sets.

This query displays German cyclists who are also German skaters at the same time:

EXCEPT

SELECT name

FROM cycling

WHERE country = 'DE'

EXCEPT / MINUSSELECT name

FROM skating

RE country = 'DE';EXCEPT returns only the rows that appear in the first result set but do n

in the second result set.

This query displays German cyclists unless they are also German skaters at the

same time:SINGLE VALUE

```
SELECT name FROM city
WHERE rating = (
  SELECT  rating
  FROM  city
  WHERE  name = 'Paris'
```

ery returns exactly one column and exactly one row. It can be used with comparison opera

This query finds cities with the same rating as Paris:

MULTIPLE VALUES

```
SELECT name
FROM city
WHERE country_id IN (
  SELECT  country_id
  FROM  country
  WHERE  population > 20000000
```

return multiple columns or multiple rows. Such subqueries can be used with operators IN

This query finds cities in countries that have a population above 20M:

CORRELATED

```
SELECT *
FROM city main_city
WHERE population > (
SELECT  AVG(population)
FROM  city average_city
```

WHERE average_city.country_id = main_city.country_id

);

This query finds countries that have at least one city:

SELECT name

FROM country

WHERE EXISTS (

SELECT *

FROM city

WHERE country_id = country.id

s introduced in the outer query. A correlated subquery depends on the outer query. It cann

This query finds cities with a population greater than the average population in the country: